

Supporting Information for "Melting in high-pressure ice layers of large ocean worlds - Implications for volatiles transport"

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Introduction

The structure of this Supporting Information is as follows: in Section S1, we provide a brief summary of the numerical model, in Section S2 we describe in detail the TBL analysis (in support of Section 2.1 in the main paper), and in Section S3 we derive the scaling law for the top temperate layer thickness H_t (in support of Section 2.3). Moreover, Figure S3 shows the root-mean-square velocities as functions of Ra (in support of Section 2.2) and Figure S4 the correlation between the outflow water velocity and the upwelling of subsolidus plumes in the convection domain (in support of Section 2.3). Finally, Table S1 provides the summary of all simulations and the corresponding data as requested by the journal so that future authors can reproduce our results.

Text S1: Governing equations and numerical implementation

Here we provide a brief summary of the numerical model, for more details see *Kalousová et al.* [2018]. We consider the HP ice layer to behave as a mixture of two components - solid ice (subscript i) and liquid water (subscript w). The mixture can exist in two distinct states: **cold ice** (temperature is below the melting point, no liquid water is present) or **temperate** (partially molten) **ice** (temperature equals the melting point and interstitial liquid water can be present). We further assume that the ice deformation does not affect the porous fluid flow, which is then driven solely by the buoyancy due to density difference between the two phases. In this setting, we solve the following equations [Souček et al., 2014]:

$$\nabla \cdot (\phi \mathbf{v}_w + (1-\phi) \mathbf{v}_i) = \frac{\Delta \rho}{\rho_w \rho_i} r, \quad (\text{S1a})$$

$$\begin{aligned} \nabla \Pi = & -\phi \Delta \rho \mathbf{g} - (1-\phi) \rho_i \alpha_i (T - T_m) \mathbf{g} + \nabla \cdot \left(\frac{(1-\phi) \mu_i}{\phi} (\nabla \cdot \mathbf{v}_i) \right) \\ & + \nabla \cdot \left((1-\phi) \mu_i \left[\nabla \mathbf{v}_i + (\nabla \mathbf{v}_i)^T - \frac{2}{3} (\nabla \cdot \mathbf{v}_i) \mathbf{I} \right] \right), \end{aligned} \quad (\text{S1b})$$

$$\mathbf{v}_w = \mathbf{v}_i - \frac{K(\phi)}{\mu_w \phi} (1-\phi) \Delta \rho \mathbf{g}, \quad (\text{S1c})$$

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}_w) = \frac{r}{\rho_w}, \quad (\text{S1d})$$

$$\rho_i c_i^p \frac{\partial T}{\partial t} + \rho_i c_i^p \mathbf{v}_i \cdot \nabla T + Lr = k_i \nabla^2 T. \quad (\text{S1e})$$

Here ϕ is the porosity (volume fraction of water in the mixture), $\mathbf{v}_w$ is the percolating water velocity, $\mathbf{v}_i$ is the convecting ice velocity, ρ_i is the ice density, ρ_w is the water density, $\Delta \rho = \rho_i - \rho_w$, r is the melting/freezing rate (positive for melting), $\Pi = P_w - P_i^{\text{ref}}$ is the excess water pressure with respect to the ice hydrostatic equilibrium pressure (P_i^{ref}), $\mathbf{g}$ is the gravity, α_i is the ice thermal expansivity, T is the ice temperature, T_m is the melting temperature, μ_i is the ice shear viscosity (cf. below), $K(\phi)$ is the porosity-dependent permeability (cf. below), μ_w is the water shear viscosity, t is the time, c_i^p is the isobaric specific heat of ice, L is the latent heat of ice, and k_i is the ice thermal conductivity. The appropriate values of all model parameters are given in detail in *Kalousová et al.* [2018]. Here, we only specify two important parameters. The **ice permeability** plays a key role in the efficiency of water extraction and generally depends on numerous parameters. For terrestrial sea ice, an abrupt permeability decrease for porosities below the ‘percolation threshold’ was reported by *Golden et al.* [1998] which corresponds to the closure of the system of veins in the ice and the subsequent drop in its connectivity. In the absence of better knowledge, we assume that the HP ice in the planetary layers behaves in a similar manner and use the following equation:

$$K(\phi) = \begin{cases} K_0 (\phi - \phi_c)^n & \phi \geq \phi_c \\ 0 & \phi < \phi_c \end{cases}, \quad (\text{S2})$$

where ϕ_c is the percolation threshold, here considered to be 1%, and K_0 , n are experimentally determined constants. The **ice viscosity** is a critical parameter that controls the dynamics of the HP ice layer. It has been measured by two groups [Sotin et al., 1985; Durham et al., 1996] and we provide a detailed comparison of their findings in *Kalousová et al.* [2018]. Both groups report that the isoviscous curves are almost parallel to the melting curve which justifies the use of the following expression:

$$\mu_i = \mu_0 \exp \left[\frac{Q}{R} \left(\frac{1}{T} - \frac{1}{T_m(P)} \right) \right], \quad (\text{S3})$$

where μ_0 is the reference viscosity at the melting temperature, Q is the activation energy, and R is the universal gas constant.

The governing equations are supplemented with the following boundary conditions: (S1a)–(S1b) free slip on top and side boundaries, no slip on the bottom boundary, and fixed excess water pressure $\Pi=0$ in one point of the computational domain, (S1c) free outflow of water at the top boundary and water-impermeable bottom and side boundaries, (S1d) no boundary condition (hyperbolic equation with no inflow boundary), (S1e) fixed temperature at the top boundary, fixed heat flux from the silicate core q_s at the bottom boundary, and insulating side boundaries. We consider a Cartesian domain of aspect ratio 2 and solve the system (S1a)–(S1e) in the open source finite element software package FEniCS [http://fenicsproject.org, Logg et al., 2012; Alnæs et al., 2015]. Details on the implementation can be found in *Kalousová et al.* [2018]. Resolution tests were

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performed in the previous study that have shown that the chosen computational mesh is sufficient to have the steady-state temperature fields well resolved and resolution independent.

Text S2: Thermal boundary layer analysis

The thermal boundary layer (TBL) analysis uses the relationship between the non-dimensional TBL thickness δ' defined as

$$\delta' = \frac{\delta}{H_c} \quad (\text{S4})$$

(here δ is the dimensional TBL thickness and H_c the thickness of the convective layer, cf. the main text for its definition) and the far field Rayleigh number Ra^+

$$Ra^+ = \frac{\alpha_i \rho_i^2 g c_i^p \Delta T_c H_c^3}{k_i \mu_i} \quad (\text{S5})$$

with ΔT_c the temperature contrast accross the convective layer (cf. main text). We can rewrite the far field Rayleigh number in terms of Roberts (convection with internal heating and specified heat flux at one of the boundaries - here q_s) or Bénard (convection with an imposed temperature contrast ΔT accross the layer and no internal heat sources) Rayleigh numbers as follows:

$$Ra^+ = Ra_R \frac{k_i \Delta T_c H_c^3}{H^4 q_s} = \frac{1}{16} Ra_B \quad (\text{S5}')$$

where Ra_R is the Roberts Rayleigh number [e.g. *Jarvis*, 1984]:

$$Ra_R = \frac{\alpha_i \rho_i^2 g c_i^p H^4 q_s}{k_i^2 \mu_i} \quad (\text{S6})$$

and Ra_B the Bénard Rayleigh number [e.g. *Jarvis*, 1984]:

$$Ra_B = \frac{\alpha_i \rho_i^2 g c_i^p \Delta T H^3}{k_i \mu_i} \quad (\text{S7})$$

Ra_B is a factor 16 larger than the far field Rayleigh number Ra^+ due to fact that the interior temperature is halfway between the bottom and top temperature and the thickness of the convective layer affecting the thermal boundary layer is half the total thickness since both boundaries play a similar role [cf. *Sotin & Labrosse*, 1999]. The advantage of using Ra^+ instead of Ra_R or Ra_B is that it is defined independently of the heating type by using the temperature difference accross the TBL ΔT_c and the convective layer thickness H_c and thus allows comparison between the configurations with internal heating, heating from below (prescribed heat flux), and constant temperature boundary conditions.

The particular form of scaling law for the TBL thickness δ' depends on the boundary conditions. In the case of rigid boundary conditions (no slip), *Mitrovica & Jarvis* [1987] give the following scaling for δ' (their eq. 13a):

$$\delta'_{ns} = 0.979 Ra_B^{-0.159} = 0.630 (Ra^+)^{-0.159} \quad (\text{S8})$$

where we already used the relation between Ra^+ and Ra_B (eq. S5').

For free slip boundary conditions, we used tabulated data from *Jarvis* [1984] (their Table 1, values of Ra_B/Ra_c in the range of 100–30 000 and the corresponding values of δ'_1) to obtain the following power-law scaling:

$$\delta'_{fs} = 1.108 (Ra^+)^{-0.303} \quad (\text{S9})$$

For the present study, we evaluated the hot TBL thickness δ from the heat flux in the hot TBL (q_δ , green lines

in Figure 1 in the main text) and the interior temperature T^i (cyan lines therein) and subtracted the thickness of the bottom temperate layer H_b in the case of regimes 1 and 2. The obtained values are provided in Table S1. Left panel of Fig. S1 shows the nondimensional TBL thickness δ' as a function of the far field Rayleigh number Ra^+ for the simulations in regime 4 (to eliminate the effect of melting on the TBL analysis). Following the approach of *Jarvis* [1984] and *Mitrovica & Jarvis* [1987], we assumed a power-law dependence and obtained the following scaling:

$$\delta' = 2.746 (Ra^+)^{-0.271} \quad (\text{S10})$$

which is indicated by the black solid line in the left panel of Fig. S1. The gray solid and dashed lines correspond to eqs (S8) and (S9), respectively, and show that the power-law coefficient of -0.271 obtained in our study is, as can be expected, between the value of -0.159 obtained if both boundaries are rigid (eq. S8) and -0.303 obtained for both being free slip (eq. S9).

The condition for no melting at the bottom boundary can be written as (cf. eqs 1 and 5 in the main text):

$$\Delta T_c = \frac{q_s \delta}{k_i} = T^s - T^i < T_m^s - T^i \quad (\text{S11})$$

Dividing both sides by H_c , using the scaling for δ' (eq. S10), and changing the inequality to equality (to obtain the critical value), we arrive at (also eq. 6 in the main text):

$$Ra_c^+ = \left(2.746 \frac{q_s H_c}{k_i (T_m^s - T^i)} \right)^{3.690} \quad (\text{S12})$$

We now compute Ra_c^+ for different HP ice layer thicknesses H , melting point viscosities μ_0 , and silicates heat fluxes q_s by the following procedure: (i) we compute the thickness of the top temperate layer H_t from the input parameters q_s and μ_0 using the scaling eq. (9) in the main text (cf. also Section S3), (ii) we approximate the interior temperature T^i by the melting temperature at the bottom of the top temperate lid, i.e. $T^i = T_m(z=H-H_t)$, (iii) we assume that there is no temperate layer at the bottom boundary ($H_b=0$) and thus $H_c=H-H_t$, and (iv) we compute Ra_c^+ using eq. (S12). The right panel of Fig. S1 shows the computed critical far field Rayleigh number Ra_c^+ as a function of heat flux from silicates q_s . Colors indicate the HP ice layer thickness H and symbols the melting point viscosity μ_0 . This plot shows that the critical far field Rayleigh number is strongly dependent on the heat flux, while it is only weakly dependent on the HP ice layer thickness and effectively independent of the melting point viscosity. We obtain the following scaling:

$$Ra_c^+ = 19.965 \times 10^3 (q_s [\text{mW m}^{-2}])^{3.690} \quad (\text{S13})$$

(black line in Fig. S1). We can also find a power-law scaling for each thickness separately (colored dashed lines in the same figure) - all with the same heat flux exponent 3.690. Dividing Ra_c^+ by $q_s^{3.690}$ (to separate the dependency on heat flux), we can find a linear dependence on the layer thickness H :

$$\frac{Ra_c^+}{(q_s [\text{mW m}^{-2}])^{3.690}} = -38.498 H [\text{km}] + 30.052 \times 10^3 \quad (\text{S14})$$

which gives the final scaling:

$$Ra_c^+ = (-38.498 H [\text{km}] + 30.052 \times 10^3) (q_s [\text{mW m}^{-2}])^{3.690} \quad (\text{S15})$$

The dependence of Ra_c^+ on H and thus $H_c/(T_m^s - T^i)$ is weak because $(T_m^s - T^i)$ increases with increasing H_c . We therefore provide only eq. (S13) in the main text (eq. 7).

Text S3: Thickness of temperate layers

The bottom temperate layer, if present (regimes 1 and 2), is very thin in comparison with the top temperate layer. Left panel of Figure S2 shows its thickness H_b as a function of the silicates heat flux q_s sorted by exchange regime (color) and viscosity (symbol). We found thicknesses of only a few kilometers that depend on the heat flux from silicates q_s as well as on the melting point viscosity μ_0 . However, given the small values of H_b , we neglect the bottom temperate layer in the analysis.

The right panel of Figure S2 shows H_t as a function of prescribed silicates heat flux q_s for different thicknesses H (colors) and viscosities μ_0 (symbols) and hints at a strong dependence of H_t on μ_0 and an important dependence on q_s . We thus plotted the top temperate layer thickness as a function of the melting point viscosity μ_0 (not shown here) and found a power-law fit of the form:

$$H_t[\text{km}] = 0.018(\mu_0[\text{Pa s}])^{0.21} \quad (\text{S16})$$

To find the dependence on heat flux, we plotted the thickness divided by $\mu_0^{0.21}$ (to separate the dependence on viscosity) as a function of heat flux q_s and we found a linear fit of the form:

$$\frac{H_t[\text{km}]}{(\mu_0[\text{Pa s}])^{0.21}} = 0.145 \times 10^{-3} q_s [\text{mW m}^{-2}] + 0.015 \quad (\text{S17})$$

Combining these two laws together, we obtain a scaling law for the top temperate layer thickness in the form:

$$H_t[\text{km}] = (0.145 \times 10^{-3} q_s [\text{mW m}^{-2}] + 0.015) \mu_0 [\text{Pa s}]^{0.21} \quad (\text{S18})$$

(also eq. 9 in the main text). Figure 4, left, in the main text shows that eq. (S18) provides a very good estimate of the observed thickness H_t .

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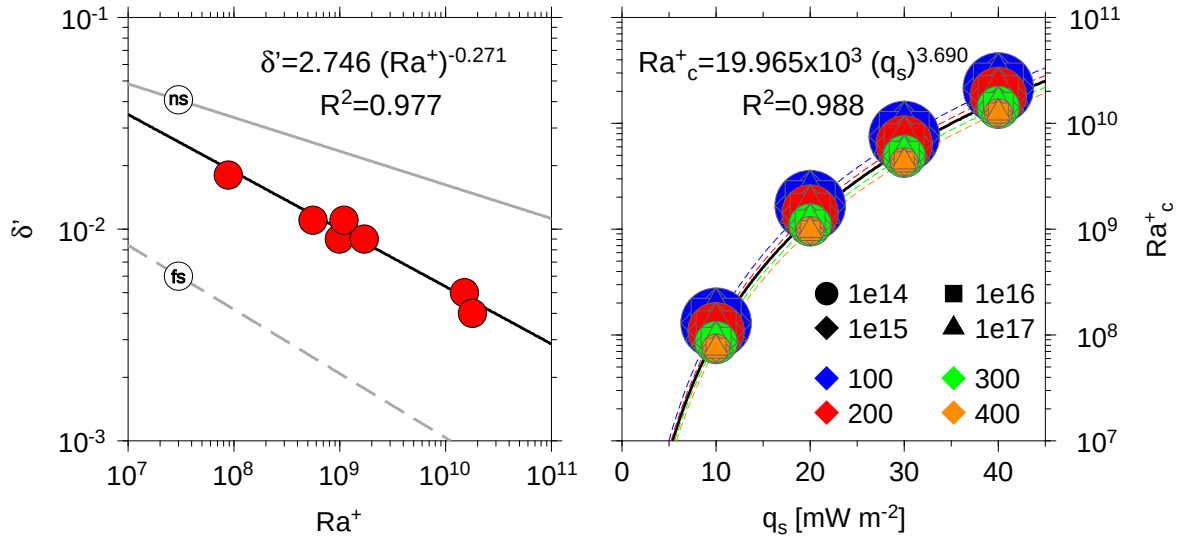


Figure S1: *Left*: Nondimensional thickness of the hot TBL (δ') as a function of the far field Rayleigh number (Ra^+) for simulations in regime 4 (red circles). Black line shows the power-law scaling (eq. S10). Gray lines indicate scaling for both boundaries being no slip (full line, eq. S8, [Mitrovica & Jarvis, 1987]) and free slip (dashed line, eq. S9, [Jarvis, 1984]). *Right*: Critical far field Rayleigh number (Ra_c^+ , eq. S12) computed for thicknesses $H=100$ – 400 km (colors), viscosities $\mu_0=10^{14}$ – 10^{17} Pa s (symbols), and silicate heat fluxes $q_s=10$ – 40 mW m^{-2} (x -coordinate). Black line shows the power-law scaling (eq. S13). Colored dashed lines correspond to power-law scalings for the particular thicknesses (eq. S15).

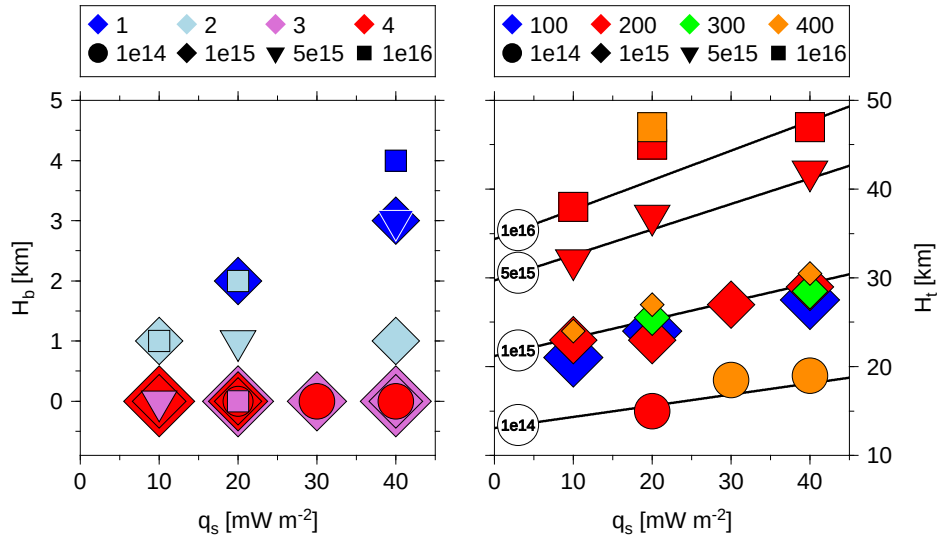


Figure S2: *Left*: Bottom temperate layer thickness H_b as a function of silicates heat flux q_s sorted by exchange regime (colors) and viscosity (symbols). *Right*: Top temperate layer thickness H_t as a function of silicates heat flux q_s sorted by layer thickness (colors) and viscosity (symbols). Lines indicate scaling eq. (S18) for different viscosities.

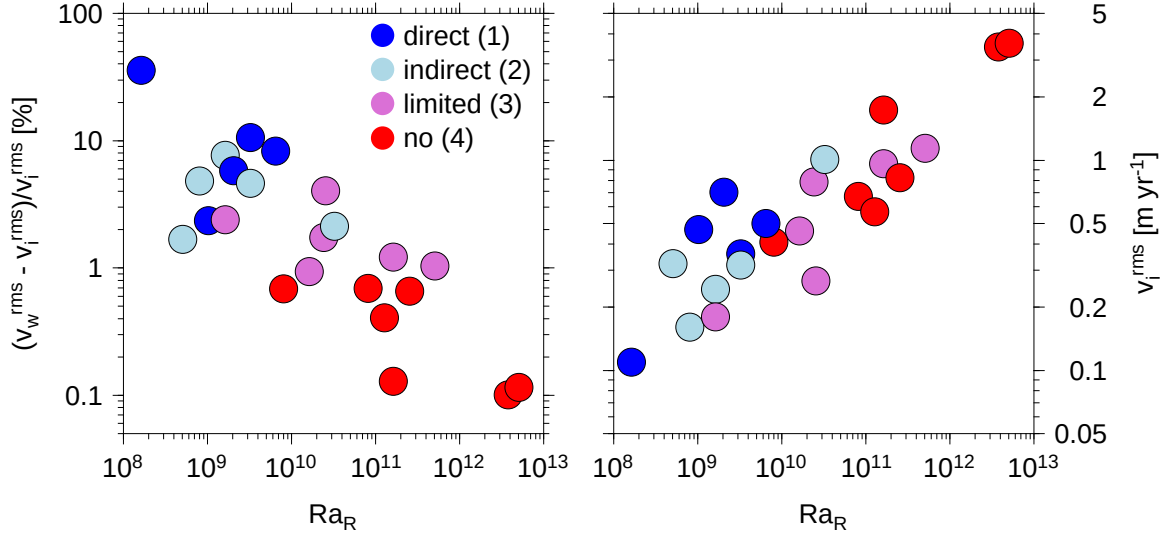


Figure S3: The relative difference of rms water velocity from the rms ice velocity, $(v_w^{\text{rms}} - v_i^{\text{rms}})/v_i^{\text{rms}}$ (left) and the rms velocity of ice (right) as a function of Roberts Rayleigh number Ra_R for all simulations. Colors indicate the exchange regime. For simulations where melting occurs at the silicates interface (regimes 1–3), the velocities range from 0.1 to 1 m yr^{-1} . Also note that with the increasing amount of melt (regimes 4 to 1), the difference between the ice and water rms velocities increases.

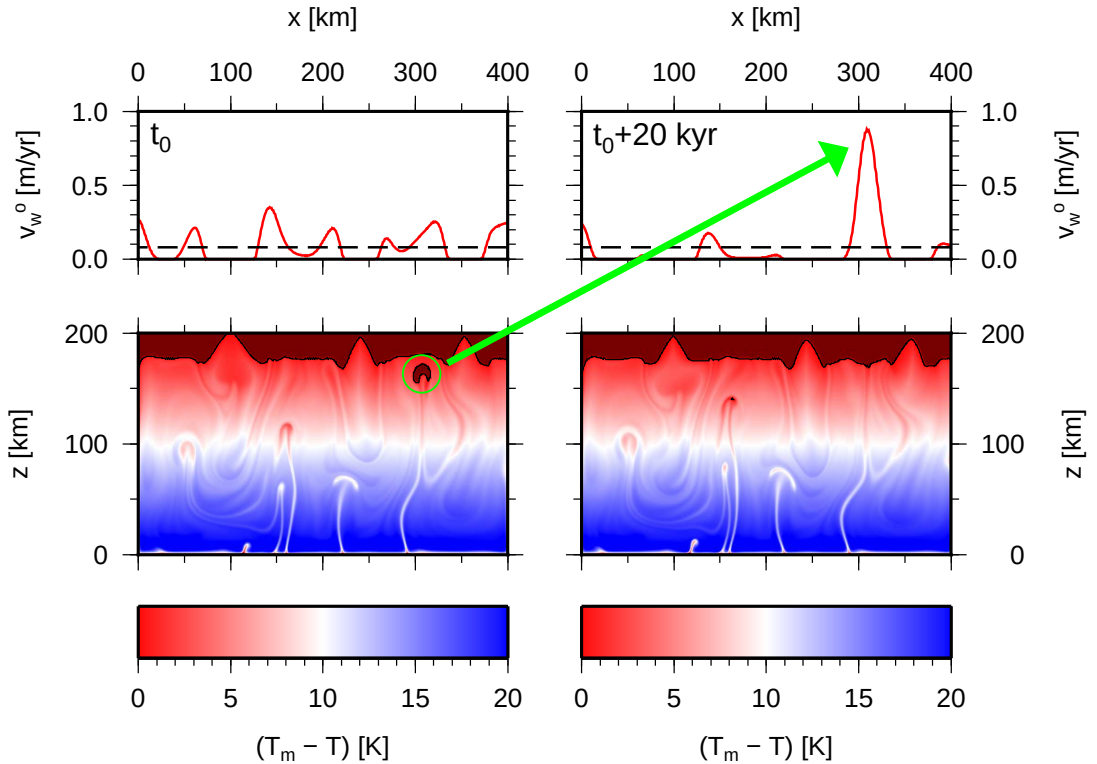


Figure S4: Reference simulation (#1 in Table S1). *Top*: outflow water velocity at the ocean interface (v_w^o) as a function of the x -coordinate. Dashed line marks the time and space average (points in the right panel of Figure 4 in the main text). *Bottom*: temperature difference from the melting temperature. Dark red regions indicate temperate ice where liquid water can exist. The time interval between the two snapshots is 20 kyr. Note the correlation between the upwelling in the temperature field (green circle) and the outflow velocity.

Table S1: List of simulations. The threshold porosity $\phi_c=1\%$. The double horizontal line marks the boundary between the original simulations from *Kalousová et al.* [2018] (#1–4, 6–16) and the new simulations designed for the purpose of the current study (#21–28).

run #	reg #	μ_0 [Pa s]	H [km]	q_s [W/m ²]	q_δ [W/m ²]	T^s [K]	T^i [K]	δ [km]	H_b [km]	H_t [km]	v_w^o [m/yr]	v_i^{rms} [m/yr]	v_w^{rms} [m/yr]	Ra _R -	Ra ⁺ -	Ra _c ⁺ -
1	3	10 ¹⁵	200	20.0	10.0	332	311	3.4	0	23	0.08	0.46	0.46	1.6×10 ¹⁰	9.5×10 ⁷	1.2×10 ⁹
2	4	10 ¹⁴	200	20.0	18.4	330	310	1.7	0	15	0.12	1.74	1.74	1.6×10 ¹¹	9.9×10 ⁸	1.3×10 ⁹
3	2	10 ¹⁶	200	20.0	7.0	332	311	5.7	2	45	0.13	0.24	0.26	1.6×10 ⁹	5.9×10 ⁶	7.8×10 ⁸
4	1	10 ¹⁷	200	20.0	2.6	332	312	14.8	6	74	0.12	0.11	0.15	1.6×10 ⁸	2.8×10 ⁵	3.6×10 ⁸
6	2	10 ¹⁵	100	10.0	5.9	332	322	2.9	1	21	0.06	0.32	0.33	5.1×10 ⁸	3.7×10 ⁶	8.2×10 ⁷
7	1	10 ¹⁵	100	20.0	6.8	332	322	2.6	2	24	0.13	0.47	0.48	1.0×10 ⁹	3.2×10 ⁶	8.6×10 ⁸
8	1	10 ¹⁵	100	40.0	6.8	332	322	2.4	3	28	0.25	0.70	0.75	2.0×10 ⁹	2.6×10 ⁶	9.6×10 ⁹
9	4	10 ¹⁵	200	10.0	9.6	330	310	3.2	0	23	0.06	0.41	0.41	8.1×10 ⁹	8.8×10 ⁷	9.0×10 ⁷
10	3	10 ¹⁵	200	30.0	11.8	332	311	3.8	0	27	0.19	0.79	0.80	2.4×10 ¹⁰	8.8×10 ⁷	5.0×10 ⁹
11	2	10 ¹⁵	200	40.0	13.2	332	311	3.0	1	29	0.27	1.01	1.03	3.2×10 ¹⁰	8.3×10 ⁷	1.4×10 ¹⁰
12	4	10 ¹⁵	300	20.0	17.2	331	297	3.2	0	26	0.12	0.67	0.68	8.1×10 ¹⁰	5.6×10 ⁸	9.5×10 ⁸
13	3	10 ¹⁵	300	40.0	18.9	332	297	3.3	0	29	0.22	0.96	0.97	1.6×10 ¹¹	5.5×10 ⁸	1.2×10 ¹⁰
14	4	10 ¹⁵	400	10.0	9.9	309	283	4.2	0	24	0.07	0.57	0.57	1.3×10 ¹¹	1.1×10 ⁹	6.7×10 ⁷
15	4	10 ¹⁵	400	20.0	19.7	325	283	3.4	0	27	0.14	0.82	0.83	2.5×10 ¹¹	1.7×10 ⁹	8.6×10 ⁸
16	3	10 ¹⁵	400	40.0	26.4	332	283	3.2	0	31	0.24	1.14	1.15	5.1×10 ¹¹	1.9×10 ⁹	1.1×10 ¹⁰
21	2	10 ¹⁶	200	10.0	6.4	332	311	5.9	1	38	0.06	0.16	0.17	8.1×10 ⁸	7.1×10 ⁶	6.7×10 ⁷
22	1	10 ¹⁶	200	40.0	7.4	332	311	5.1	4	47	0.25	0.36	0.40	3.2×10 ⁹	5.5×10 ⁶	8.8×10 ⁹
23	3	5×10 ¹⁵	200	10.0	6.7	332	311	5.3	0	32	0.06	0.18	0.18	1.6×10 ⁹	1.6×10 ⁷	7.7×10 ⁷
24	2	5×10 ¹⁵	200	20.0	8.6	332	311	5.1	1	37	0.13	0.32	0.33	3.2×10 ⁹	1.4×10 ⁷	9.1×10 ⁸
25	1	5×10 ¹⁵	200	40.0	9.0	332	311	4.5	3	42	0.27	0.50	0.54	6.5×10 ⁹	1.3×10 ⁷	9.7×10 ⁹
26	3	10 ¹⁶	400	20.0	12.9	332	284	6.2	0	47	0.13	0.27	0.28	2.5×10 ¹⁰	1.7×10 ⁸	7.6×10 ⁸
27	4	10 ¹⁴	400	30.0	29.7	317	282	1.8	0	19	0.22	3.46	3.46	3.8×10 ¹²	1.5×10 ¹⁰	4.0×10 ⁹
28	4	10 ¹⁴	400	40.0	39.2	323	282	1.7	0	19	0.27	3.61	3.61	5.1×10 ¹²	1.8×10 ¹⁰	1.2×10 ¹⁰